

Abhishek Singh

CS undergrad at KIIT with hands-on experience building and deploying full-stack AI applications using Python, FastAPI, React, and Docker.

[gmail](#) | [portfolio](#) | [linkedin](#) | [github](#) | [itch.io](#)

EDUCATION

Kalinga Institute of Industrial Technology (KIIT)

Bachelor of Technology in Computer Science and Engineering

Bhubaneswar, India

2023 – Present | CGPA: 8.48

PROJECTS

Talk with CSV | *React, FastAPI, Groq AI (Llama 3.3), Python, Pandas, Matplotlib* | [GitHub](#) 2026

- Built a full-stack AI web app that lets users upload any CSV and ask questions about it in plain English
- Designed a FastAPI backend that parses CSVs with Pandas, sends schema + query to Groq's Llama 3.3-70B, and returns structured JSON with answers, chart specs, and tabular results
- Implemented server-side chart generation (bar, line, pie, scatter, histogram) using Matplotlib with smart aggregation logic (mean/sum/count) inferred from the question
- Built a React frontend with session-based file handling, inline chart rendering, suggestion chips, and a typing indicator
- Architected a clean separation between AI prompt engineering, data processing, and UI rendering layers

Disease Risk Predictor | *Python, Scikit-learn, Streamlit, Docker* | [Live](#) | [GitHub](#) 2026

- Built an end-to-end ML web app predicting Diabetes and Heart Disease risk from real medical datasets
- Applied SMOTE for class imbalance, engineered interaction features, and ran GridSearchCV across 405 model fits
- Compared Logistic Regression, Random Forest, Gradient Boosting, and SVM; achieved 92.44% accuracy on heart disease
- Saved trained models and scalers as .pkl files; applied consistent StandardScaler transformation at inference
- Containerized with Docker and deployed live on Render

Financial Transaction Fraud Detection System | *Python, Scikit-learn, FastAPI, Streamlit, Docker* | [GitHub](#) 2026

- Developed an end-to-end ML system to classify financial transactions as fraudulent or non-fraudulent
- Trained a Random Forest model on transaction features such as type, amount, and balance changes
- Built a REST API using FastAPI for real-time predictions; designed interactive Streamlit frontend
- Containerized with Docker for consistent deployment

Game Development – IIT Guwahati Game Jam (GameDeon) | *Godot, GDScript* | [itch.io](#) 2025

- Secured **7th position among 50 teams** nationwide; invited to IIT Guwahati for final showcase
- Built and shipped a complete game as part of a 2-member team under strict time constraints

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, R

Frontend: React, HTML, CSS

Backend & APIs: FastAPI, REST API design, Session management

AI & LLMs: Groq API, LLM prompt engineering, Structured output parsing

Data Science: Pandas, NumPy, Matplotlib, Seaborn, EDA, Feature Engineering

ML: Scikit-learn, Random Forest, Gradient Boosting, Logistic Regression, SVM, SMOTE, GridSearchCV, TF-IDF

Databases: SQL, MariaDB, SQLite

Dev Tools: Git, GitHub, Docker, VS Code, Jupyter Notebook, Render, Linux (CLI)

Other: Streamlit, Godot Engine, GitHub Pages

CERTIFICATIONS

IBM Data Science Professional Certificate Coursera

Courses: What is Data Science, Tools for Data Science, Data Science Methodology, Python for Data Science

Google: Foundations: Data, Data, Everywhere Coursera

Explored data-driven decision-making, data collection, analysis, and interpretation

ADDITIONAL

Competitive Programming

Regularly practice DSA on LeetCode and CodeChef (Python, C++); focus on arrays, linked lists, graphs, and DP